

AutoMatik

Integrated Car Dealership & Management System

UI / UX Requirements Document

Version 1.0 · Academic Year 2025–2026

Prepared for:	Polytechnic University of the Philippines — CCIS
System:	AutoMatik Web-Based Car Dealership & Management System
Document Type:	UI / UX Requirements — Frontend Scope
Portals:	Landing Page · Admin Portal · Agent Portal · Customer Portal
Tech Stack:	React + Vite + Tailwind CSS (API: Flask / Python)
Date:	June 2026

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1. Purpose & Scope

This document defines the complete User Interface (UI) and User Experience (UX) requirements for AutoMatik — a web-based car dealership and management system. It covers all four user-facing portals: Landing Page, Admin Portal, Agent Portal, and Customer Portal.

Each section specifies sidebar navigation menus, page-level content, data fields displayed or captured, and supporting UI components derived directly from the Flask API flow and MySQL database schema.

1.1 Document Scope

Portal	Primary Users	Access Method
Landing	Public / Guest	Open URL — no login required
Admin	Administrator/Manager	Username + password → session cookie
Agent	Sales Agent	Username + password → session cookie
Customer	Registered Customer	Token-based portal link (post-purchase)

Note: Customers receive portal access only after a completed purchase. Self-registration is not permitted per system constraints.

2. General UI Conventions

2.1 Layout

- Two-column layout: fixed left sidebar (240 px) + main content area.
- Top navigation bar: system logo, portal name, notification bell, user avatar with dropdown.
- Responsive breakpoints: Desktop (≥ 1280 px), Tablet (768–1279 px), Mobile (< 768 px).
- Sidebar collapses to icon-only mode on tablet; bottom sheet on mobile.

2.2 Typography & Colors

- Primary font: Calibri (web fallback: Helvetica Neue, sans-serif).
- Base body size: 14 px / Line-height: 1.5.
- Portal accent colors to differentiate role context (Admin: dark navy, Agent: steel blue, Customer: teal).
- Danger/Error: #E74C3C · Success: #27AE60 · Warning: #F39C12 · Info: #3498DB.

2.3 Forms & Validation

- All required fields marked with an asterisk (*).
- Inline validation — error message appears below field on blur.
- Disabled submit button until all required fields pass validation.
- Date pickers use ISO 8601 format (YYYY-MM-DD) internally; display as MM/DD/YYYY.
- Currency fields formatted with comma separators and two decimal places (e.g., ■ 1,234,567.00).

2.4 Tables & Lists

- All data tables include: search bar, column sort, pagination (10/25/50 rows per page), and export (CSV/PDF).
- Status badges: color-coded chips (e.g., pending = gray, active = green, overdue = red).
- Row actions: Edit, View, Delete — shown as icon buttons in last column.
- Empty state: illustrated empty-state card with a call-to-action button.

2.5 Notifications & Feedback

- Toast notifications (top-right, auto-dismiss 4 s) for CRUD success/failure.
- Modal confirmations for irreversible actions (delete, cancel, mark paid).
- Loading skeletons for data-heavy pages (tables, dashboard cards).
- Notification bell icon shows unread count badge; dropdown shows last 5 alerts.

3. Landing Page

LANDING PORTAL

Public-facing marketing site. No authentication required. Entry point for guest inquiries, vehicle browsing, chatbot, service slot viewing, and the payment calculator tool.

3.1 Navigation Bar (Top)

Nav Item	Destination / Behavior
Logo	Scrolls to Hero section
Home	Hero / banner section
Vehicles	Vehicle listings section / page
About	Dealership info section
Contact / Inquiry	Inquiry Form section
Services	Service booking info + slot calendar
Login	Modal or redirect: /auth/login

3.2 Page Sections & Content

Section	Content Description	Data Source / API
Hero Banner	Full-width banner, headline, CTA buttons: 'Browse Vehicles', 'Get a Quote'	Static / CMS
Featured Vehicles	Grid of 4–6 vehicles: photo, brand, model, year, price, 'View Details' button	GET /vehicles (status=available)
Vehicle Detail	Full specs, photo carousel, inquiry button, payment calculator widget	GET /vehicles/
Inquiry Form	Name*, email*, phone, vehicle selection dropdown, message textarea	POST /inquiries
Payment Calculator	Vehicle price, down payment, term (months), interest rate → monthly estimate	Client-side formula (no API call)
Service Slots	Calendar/list of available test drive slots; click to pre-fill booking form	GET /service/slots
Chatbot Widget	Floating button (bottom-right); opens chat drawer with AI responses	POST /chatbot/message
About / Contact	Dealership address, hours, map embed, social links	Static
Footer	Nav links, privacy policy, terms, copyright	Static

3.3 Inquiry Form — Data Fields

Field	Type	Required	DB Column
Full Name	Text	Yes	inquiries.guest_name
Email Address	Email	Yes	inquiries.guest_email
Phone Number	Tel	No	inquiries.guest_number
Vehicle Interest	Dropdown	Yes	inquiries.vehicle_id
Message	Textarea	Yes	inquiries.message

4. Admin Portal

ADMIN PORTAL

Full system access. Manages all entities: users, vehicles, sales, loans, payments, documents, warranty, bookings, suppliers, audit logs, and settings.

4.0 Sidebar Navigation

Menu Item	Icon (suggestion)	Sub-items
Dashboard	grid	—
Users	users	All Users · Agents · Customers
Vehicles	car	Vehicle List · Add Vehicle · Low Stock
Suppliers	truck	Supplier List · Supplies / Parts
Inquiries	message-circle	All Inquiries · Chatbot Logs
Sales	shopping-cart	All Sales · Contracts · Insurance
Loans	credit-card	Loan List · Amortization · Overdue
Payments	dollar-sign	Payment List · Summary
Commissions	trending-up	Commission List · Agent Performance
Service Bookings	calendar	Bookings · Slots Management
Warranty Claims	shield	Claim List
Documents	file-text	Document List
Notifications	bell	Send Notification
Audit Logs	activity	—
Settings	settings	System Settings

4.1 Dashboard

KPI Card	Value Source	Drill-down Target
Total Vehicles Available	COUNT vehicles WHERE status=available	Vehicle List
Total Sales (Month)	COUNT sales WHERE MONTH(sale_date)=current	Sales List
Revenue (Month)	SUM selling_price WHERE month=current	Sales Summary
Pending Inquiries	COUNT inquiries WHERE status=open	Inquiry List

KPI Card	Value Source	Drill-down Target
Overdue Amortizations	COUNT amortization WHERE status=overdue	Amortization Overdue
Unpaid Commissions	COUNT commissions WHERE is_paid=0	Commission List
Pending Warranty Claims	COUNT warranty WHERE status=submitted	Warranty List

- Recent Activity Feed: last 10 audit_logs with actor name, action, table, timestamp.
- Sales Chart: Bar chart — monthly revenue for last 6 months (data from sales JOIN amortization).
- Top Agents Widget: ranked by commission_amount DESC for current month.

4.2 User Management

- User List Table: user_id, full_name, email, role badge, is_active toggle, last_login, actions.
- Create/Edit User Modal: username, email, password, role selector, profile fields.
- Agent Detail: employee_number, hire_date, default_commission_rate, total sales count.
- Customer Detail: customer_number, preferred_contact, preferred_payment, sales history list.

4.3 Vehicle & Inventory Management

Field	Type	DB Column
Brand	Text	vehicles.brand
Model	Text	vehicles.model
Year	Number	vehicles.year
Color	Text	vehicles.color
Body Type	Text	vehicles.body_type
Transmission	Select	vehicles.transmission
Fuel Type	Select	vehicles.fuel_type
Price (■)	Decimal	vehicles.price
Status	Select	vehicles.status
Specs (JSON)	Textarea	vehicles.specs_json
Photos	File Upload	vehicle_photos.photo_url
Supplier	Dropdown	vehicles.supplier_id → suppliers.company_name

- Status transition controls: available → reserved → delivered, any → discontinued.
- Low-stock alert badge: shown when available count < threshold (from system_settings).

4.4 Inquiries

- List table: inquiry_id, guest_name / user full_name, vehicle, status badge, created_at, assigned agent.
- Detail drawer: message thread, vehicle info card, assign agent dropdown, status action buttons.

- Chatbot Logs sub-page: session_id, user_message, bot_response, intent_tag, timestamp.

4.5 Sales & Contracts

Field	Type	DB Column
Vehicle	Dropdown	sales.vehicle_id (status=available)
Customer	Dropdown	sales.customer_id
Agent	Dropdown	sales.agent_id
Selling Price	Decimal	sales.selling_price
Payment Type	Select	sales.payment_type (cash / installment)
Sale Date	Date	sales.sale_date
Inquiry Link	Dropdown	sales.inquiry_id (optional)
Contract URL	File/URL	sales_contracts.contract_url
Contract Status	Select	sales_contracts.status
Insurance Provider	Text	insurance_records.provider_name
Policy Number	Text	insurance_records.policy_number
Coverage Type	Text	insurance_records.coverage_type

4.6 Loans & Amortization

Field	Type	DB Column
Down Payment (■)	Decimal	loan_details.down_payment
Loan Amount (■)	Decimal	loan_details.loan_amount
Interest Rate (%)	Decimal	loan_details.interest_rate
Term (months)	Number	loan_details.term_months
Monthly Amort.	Computed	loan_details.monthly_amortization (auto-calc)
Bank Name	Text	loan_details.bank_name
Bank Approval	Select	loan_details.bank_approval_status

- Amortization Schedule Table: month #, due_date, principal, interest, total_due, running_balance, status badge, mark-paid button.
- Overdue sub-page: filtered view of status=overdue rows with customer and sale context.

4.7 Payments

Field	Type	DB Column
Sale	Dropdown	payments.sale_id

Field	Type	DB Column
Schedule Row	Dropdown	payments.schedule_id (if installment)
Amount Paid (■)	Decimal	payments.amount_paid
Payment Method	Select	payments.payment_method
Reference #	Text	payments.reference
Recorded By	Auto	payments.recorded_by (session user)
Payment Date	Date	payments.payment_date

- Payment Summary card: total collected, total pending, total overdue — grouped by month.

4.8 Service Bookings & Slots

- Slots Management: create/edit slots with slot_datetime, slot_type (test_drive / maintenance / repair), capacity, is_available toggle.
- Bookings List: booking_id, customer name, vehicle, slot datetime, booking_type, warranty_claim_id (if any), status badge, confirm/complete action buttons.
- Calendar view (optional enhancement): visual grid of slots with booking count per slot.

4.9 Warranty Claims

- List table: claim_id, customer name, vehicle, claim_type badge, status badge, submitted_at, reviewed_by.
- Detail page: description text, status action buttons (under_review → approved / rejected), resolution textarea, linked service_bookings.

4.10 Documents

Document Type	Enum Value	Description
Official Receipt	OR	Cash payment proof
Certificate of Reg.	CR	LTO registration document
Warranty Certificate	warranty_cert	Vehicle warranty PDF
Amortization PDF	amortization_schedule	Generated payment schedule
Sales Contract	sales_contract	Signed contract PDF
Other	other	Miscellaneous documents

- Upload: file_url input, document_type select, sale linkage, is_accessible toggle.

4.11 Suppliers & Supplies

- Supplier List: company_name, contact_name, contact_email, contact_phone, is_active toggle.
- Supplier Detail: supplier info + linked vehicles count + supplies/parts list.

- Supplies List: part_name, part_number, unit_cost, stock_qty, reorder_level, supplier name, low-stock alert row highlight.

4.12 Notifications

- Broadcast form: select target (user_id or role), title, message, channel (in_app / email / sms).
- Auto-trigger events displayed in a read-only log for admin reference.

4.13 Audit Logs

- Table: log_id, user (actor), action badge (INSERT/UPDATE/DELETE), table_name, record_id, ip_address, created_at.
- Detail panel: old_value JSON vs new_value JSON diff viewer (side-by-side or unified diff).
- Filters: action type, table_name, user_id, date range. Paginated (50 rows default).

4.14 System Settings

Key	Description	Input Type
default_commission_rate	Fallback agent commission rate (%)	Decimal
chatbot_enabled	Toggle AI chatbot endpoint on/off	Boolean Toggle
max_loan_term_months	Cap for loan term validation	Number

- Each setting row: key (read-only), current value (editable), description, last updated by, updated_at timestamp.

5. Agent Portal

AGENT PORTAL

Sales agent workspace. Focused on managing assigned inquiries, tracking tasks, monitoring personal commissions, and handling customer follow-ups.

5.0 Sidebar Navigation

Menu Item	Sub-items
Dashboard	—
Inquiries	Assigned Inquiries
Tasks	My Tasks · Create Task
Commissions	My Earnings
Profile	View / Edit Profile

5.1 Agent Dashboard

Widget	Data Source
Assigned Inquiries Count	COUNT inquiries WHERE agent_id=me AND status=assigned
Pending Tasks	COUNT agent_tasks WHERE agent_id=me AND status=pending
Total Commissions (Month)	SUM commission_amount WHERE agent_id=me AND month=current
Unpaid Commissions	SUM commission_amount WHERE agent_id=me AND is_paid=0
Recent Sales	SELECT sales WHERE agent_id=me ORDER BY sale_date DESC LIMIT 5

5.2 Agent Inquiries

- List table: inquiry_id, customer/guest name, vehicle, message preview, status badge, created_at.
- Detail drawer: full message, vehicle info card, customer contact details, status action (resolve), linked tasks.
- Resolve button: sets status=resolved, resolved_at=NOW().

5.3 Agent Tasks

Field	Type	DB Column
Task Type	Select	agent_tasks.task_type (follow_up / appointment / demo / document_prep / other)
Title	Text	agent_tasks.title

Field	Type	DB Column
Due Date	DateTime	agent_tasks.due_date
Status	Select	agent_tasks.status
Linked Inquiry	Dropdown	agent_tasks.inquiry_id (optional)
Notes	Textarea	agent_tasks.notes

- Kanban board view (optional): columns for pending / in_progress / done / cancelled.

5.4 Agent Commissions

- List table: commission_id, sale vehicle (brand + model), selling_price, rate_applied (%), commission_amount, is_paid badge, paid_at.
- Summary card: total earned (all time), total paid, total pending.

Note: Agents view only; marking commissions paid is an Admin-only action.

6. Customer Portal

CUSTOMER PORTAL

Post-purchase self-service portal. Customers access their sales, loan schedules, documents, service bookings, warranty claims, and inquiries. Access is granted via a token-based link issued after a completed purchase.

6.0 Sidebar Navigation

Menu Item	Sub-items
Dashboard	—
Vehicles	Browse Available Vehicles
My Sales	Sale Details · Contract · Insurance
Payments	Payment History · Amortization Schedule
Documents	OR, CR, Warranty Cert, Contracts
Service	My Bookings · Book a Service
Warranty	My Claims · Submit Claim
Inquiries	My Inquiries · New Inquiry
Notifications	Inbox
Profile	Edit Profile · Preferences

6.1 Customer Dashboard

Widget	Data Source
Next Payment Due	MIN(amortization.due_date) WHERE status=unpaid AND loan=mine
Next Payment Amount	amortization.total_due for that row
Active Sales Count	COUNT sales WHERE customer_id=me AND status=active
Open Inquiries	COUNT inquiries WHERE user_id=me AND status!=closed
Unread Notifications	COUNT notifications WHERE user_id=me AND is_read=0
Recent Documents	SELECT documents WHERE sale.customer_id=me ORDER BY created_at DESC LIMIT 3

6.2 Vehicle Browse

- Same public vehicle listing: grid cards with photo, brand, model, year, price, fuel type.

- Filters: brand, model, fuel_type, transmission, price range slider.
- Vehicle detail page: full photo carousel, all specs_json fields, inquiry button.

6.3 My Sales & Contracts

- Sales list: vehicle photo, brand + model, sale_date, selling_price, payment_type badge, sale status badge.
- Sale detail accordion: contract status + download link, insurance info card, linked documents.

6.4 Loans & Amortization

- Loan summary card: loan_amount, interest_rate, term_months, monthly_amortization, bank_name, bank_approval_status badge.
- Amortization table: month #, due_date, principal, interest, total_due, running_balance, status badge (color-coded).
- Progress bar: % of total loan paid vs remaining.

6.5 Payment History

- Table: payment_date, amount_paid, payment_method badge, reference number, linked schedule month.
- Summary totals: total paid, total remaining balance.

6.6 My Documents

- Grid of document cards: icon by document_type, label, date uploaded, 'View / Download' button.
- Only shows is_accessible=1 records.
- Document types: OR, CR, warranty_cert, amortization_schedule, sales_contract, other.

6.7 Service Bookings

Field	Type	DB Column / Source
Slot	Calendar/Dropdown	service_slots WHERE is_available=1
Booking Type	Select	service_bookings.booking_type (test_drive / maintenance / repair)
Vehicle	Dropdown	service_bookings.vehicle_id (customer's purchased vehicle)
Notes	Textarea	service_bookings.notes
Warranty Claim	Dropdown	service_bookings.warranty_claim_id (optional, if repair)

- Booking list: slot datetime, booking_type, vehicle, status badge, cancel button (if pending).

6.8 Warranty Claims

Field	Type	DB Column
Sale	Dropdown	warranty_claims.sale_id (customer's sales)

Field	Type	DB Column
Vehicle	Auto-fill	warranty_claims.vehicle_id (from sale)
Claim Type	Select	warranty_claims.claim_type (repair / replacement / refund)
Description	Textarea	warranty_claims.description

- Claims list: claim_id, claim_type badge, status badge, submitted_at, resolution (if resolved).

6.9 My Inquiries

- Inquiry list: vehicle, message preview, status badge, created_at, assigned agent name.
- Submit new inquiry: same form as landing page inquiry, vehicle pre-selectable.

6.10 Notifications

- Notification list: title, message, channel badge (in_app / email / sms), is_read indicator, created_at, mark-read button.
- Mark all read button in list header.

6.11 Profile Settings

Field	Type	DB Column
Full Name	Text	user_profile.full_name
Phone Number	Tel	user_profile.phone_number
Address	Text	user_profile.address
City / Province / ZIP	Text	user_profile.city / province / zip_code
Date of Birth	Date	user_profile.date_of_birth
Gender	Select	user_profile.gender
Profile Picture	File Upload	user_profile.profile_picture_url
Preferred Contact	Select	customer_details.preferred_contact_method
Preferred Payment	Select	customer_details.preferred_payment_method
Change Password	Password	users.hash_password (via PUT /profile/password)

7. Global Components

7.1 Authentication Screens

- Login Page: email + password fields, 'Forgot Password' link, submit button. Redirect by role after login.
- Forgot Password: email input → trigger POST /auth/forgot-password → success message.
- Reset Password: new password + confirm password → POST /auth/reset-password (token from URL param).
- Email Verification: landing page for /auth/verify-email?token=... → success/error state.

7.2 Error & Empty States

- 404 Page: 'Page not found' with back-to-dashboard button.
- 403 Page: 'Access denied' — shown when role check fails.
- Empty Tables: illustrated empty state with action CTA per context.
- API Error Toast: shown on 4xx/5xx responses with short message.

7.3 Shared UI Components

Component	Usage
DataTable	All list views — search, sort, paginate, export
StatusBadge	Colored chip for enum status fields across all tables
ConfirmModal	Destructive action confirmation (delete, cancel, paid)
FileUpload	Document and photo upload with preview
CurrencyInput	■-prefixed decimal input with comma formatting
DatePicker	ISO date picker for all date/datetime fields
NotificationBell	Top-bar bell with unread count badge + dropdown
AvatarDropdown	User avatar, name, role, logout option
SidebarNav	Collapsible role-specific navigation
SearchableDropdown	Vehicle, customer, agent, inquiry selectors
JsonDiffViewer	Audit logs old vs new value side-by-side
AmortizationTable	Month-by-month breakdown with status indicators
PaymentCalculator	Client-side installment estimation widget
ChatbotWidget	Floating chat drawer (landing + logged-in views)

8. Data Field Reference by Portal

Quick-reference table mapping every primary data entity to its portal visibility, showing which fields are displayed vs editable per role.

Entity	Admin	Agent	Customer	Public
users	View + Edit	Own only	Own only	—
user_profile	View + Edit all	Own only	Own only	—
agent_details	View + Edit	View own	—	—
customer_details	View + Edit	—	View + Edit own	—
vehicles	Full CRUD	—	View only	View only
vehicle_photos	Upload + Delete	—	—	View only
suppliers	Full CRUD	—	—	—
supplies	Full CRUD	—	—	—
inquiries	Full CRUD + Assign	Own assigned	Own only	Submit only
chatbot_logs	View all	—	Own session	Own session
sales	Full CRUD	—	View own	—
sales_contracts	Full CRUD	View assigned	View own	—
loan_details	Full CRUD	—	View own	—
amortization_schedule	View + Edit	—	View own	—
payments	Record + View	—	View own	—
agent_commissions	View + Mark Paid	View own	—	—
agent_tasks	—	Full CRUD own	—	—
documents	Full CRUD	View linked	View own (accessible)	—
insurance_records	Full CRUD	—	View own	—
service_slots	Full CRUD	—	—	View available
service_bookings	View + Confirm	—	Create + Cancel own	—
warranty_claims	Review + Resolve	—	Submit + View own	—
notifications	Broadcast	Own only	Own only	—
audit_logs	View all	—	—	—
system_settings	Full CRUD	—	—	—

Note: This document is to be used as the primary specification handoff from the SA/DB teams to the UI/UX and FS (Frontend Systems) teams. Any deviations require written approval and must be reflected in the next document revision.